

SOFTWARE PROJECT DOCUMENT

Project Title: Appointment Management System

1. Introduction

1.1 Background of the Project

In many organizations, appointment booking is handled manually using paper, phone calls, or informal communication. This traditional approach often leads to problems such as double booking, loss of records, delays, and inefficiency. With the growth of technology, there is a need for a computerized system to manage appointments effectively.

1.2 Problem Statement

Manual appointment systems are inefficient, time-consuming, and error-prone. Users may face difficulties in booking appointments, while administrators struggle to manage schedules properly. These challenges reduce productivity and user satisfaction.

1.3 Proposed Solution

The proposed solution is an Appointment Management System that allows users to book appointments electronically and enables administrators to manage and monitor appointments efficiently.

2. Objectives of the Project

General Objective: To design and develop a web-based Appointment Management System that simplifies appointment booking and management.

Specific Objectives:

- Allow users to book appointments online
- Enable administrators to manage appointments
- Reduce appointment conflicts
- Improve efficiency and data organization

3. Scope of the Project

The system includes user registration, appointment booking, viewing appointments, and basic admin management. Advanced features such as payment and notifications are excluded at this stage.

4. Definition of the System

An Appointment Management System is a software application that enables users to schedule, manage, and track appointments electronically.

5. System Users

User: Books and views appointments

Admin: Manages appointments and users

6. Functional Requirements

- User registration and login
- Appointment booking
- Admin approval or rejection of appointments

7. Non-Functional Requirements

- Security
- Performance
- Usability
- Scalability

8. System Architecture

The system follows a three-tier architecture: Frontend, Backend, and Database.

9. Technologies Used

- JavaScript / TypeScript
- Next.js
- NestJS
- PostgreSQL

10. Development Methodology

The project follows an incremental development approach, building features step by step.

11. Conclusion

The Appointment Management System automates appointment booking, reduces errors, and improves efficiency.